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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/567,630	12/11/2014	Cynthia Richard	DG-001-CIP-US	4634

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EXAMINER
LU, FRANK WEI MIN

ART UNIT	PAPER NUMBER
1634	

NOTIFICATION DATE	DELIVERY MODE
01/26/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 14/567,630	Applicant(s) RICHARD, CYNTHIA	
	Examiner FRANK LU	Art Unit 1634	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE *3* MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☒ Responsive to communication(s) filed on *9/1/2017*.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL**. 2b)☒ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5)☒ Claim(s) *1,3-7 and 9-21* is/are pending in the application.

5a) Of the above claim(s) *7,9 and 11* is/are withdrawn from consideration.

6)☐ Claim(s) _____ is/are allowed.

7)☒ Claim(s) *1,3-6,10 and 12-21* is/are rejected.

8)☐ Claim(s) _____ is/are objected to.

9)☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☒ The drawing(s) filed on *12/30/2014* is/are: a)☒ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. _____ .

4) ☐ Other: _____.

DETAILED ACTION

CONTINUED EXAMINATION UNDER 37 CFR 1.114 AFTER FINAL REJECTION

1. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission of RCE and the amendment filed on September 1, 2017 have been entered. The claims pending in this application are claims 1, 3-7, and 9-21 wherein claims 7, 9, and 11 have been withdrawn due to the restriction requirement mailed on December 22, 2016. Rejection and/or objection not reiterated from the previous office action are hereby withdrawn in view of applicant's amendment filed on September 1, 2017. Claims 1, 3-6, 10, and 12-21 will be examined.

Claim Rejections - 35 USC § 112

2. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

3. Claims 5, 12, 14, and 16 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

4. Claim 5 is rejected as vague and indefinite because it is unclear what is the relationship between blunt ended duplex in the claim and the adaptor-ligated target sequence in claim 1. Please clarify.

Claim Rejections - 35 USC § 103

5. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

6. Claims 1, 3, 4, 6, 10, 13, 15, 17, 18, and 20 are rejected under 35 U.S.C. 103 as being unpatentable over Shapero *et al.,* (US 2012/0010087 A1, published on January 12, 2012) in view of Bradbury *et al.,* (US 2003/0207282 A1, published on November 6, 2003).

Regarding claims 1, 3, 4, 6, 10, and 20, Shapero *et al.,* teach a method for ligating an adapter to a 5' end in a population of nucleic acids, wherein at least some of the nucleic acids in the population of nucleic acids contain the target sequence, the method comprising the following steps, performed in order: (a) combining the population of nucleic acids (eg., target fragments of a population of nucleic acid molecules) and a target isolation probe (eg., one of a population of capture probes) in a solution, wherein the target isolation probe is a nucleic acid probe and comprises an affinity binding domain; (b) permitting a single stranded region of the target isolation probe to hybridize to all or a portion of a target sequence in the population of nucleic acids, thereby producing hybridized nucleic acids comprising a 5' single stranded target sequence; (c) enriching the hybridized nucleic acids of step (b) by associating the target isolation

probe with a capture domain (eg., by hybridizing to a nucleic acid probe attended to a solid support or biotin-streptavidin interaction), thereby producing enriched nucleic acids comprising the hybridized nucleic acids; and (d) extending, using a polymerase, the 3' end of the target isolation probe in the enriched nucleic acids using the 5' single stranded target sequence as a template to form a blunt end or a staggered end at the 5' end of the target sequence as recited in claim 1 wherein the portion of the target sequence of (b) is positioned at the 3' end of the target sequence as recited in claim 3, the affinity binding domain is positioned between the 3' end and the 5' end of the target isolation probe as recited in claim 4, the extension step (d) is performed in the presence of a modified deoxynucleotide (eg., a labeled dNTP or a labeled ddNTP) as recited in claim 6, the target isolation probe (eg., one of a population of capture probes) is modified wherein the modification is selected from an inverted base, a carbon linker, a phosphorothioate linkage or a dideoxynucleotide as recited in claim 10, and quantifying and/or sequencing the target sequence (eg., by sequencing at least a portion of the amplified target molecules) as recited in claim 20 (see abstract, paragraphs [0012], [0019], [0026], [0040] to [0042], [0080], [0082], [0083], [0101], [0103], and [0151], Figures 2 and 5-7, and claims 20, 26 and 29).

Shapero *et al.,* do not disclose ligating a 5' adaptor to the 5' end of the target sequence at the blunt or staggered end of step (d) to produce an adaptor-ligated target sequence as recited in claim 1. However, Shapero *et al.,* teach that the 5' adaptor is a duplex adaptor or a hairpin adaptor as recited in claim 13, the 5' adaptor is a next generation sequencing (NGS) platform-specific adaptor, an adaptor that contains a single nucleotide overhang, a Y structure or hairpin adaptor with a cleavable site, a fully complementary double-stranded DNA (dsDNA) adaptor, or

a dsDNA adaptor with a single-stranded DNA (ssDNA) overhang at the opposite end from the ligation junction as recited in claim 15, and the 5' adaptor comprises at least one of a sequencing primer site, a library amplification primer site (eg., a first common priming sequence or T7 promoter sequence), a unique sample identifier and a unique molecule identifier sequence as recited in claim 17 (see abstract, paragraphs [0012], [0019], [0026], [0040] to [0042], [0080], [0082], [0083], [0101], [0103], and [0151], Figures 2 and 5-7, and claims 20, 26 and 29).

Bradbury *et al.,* teach to produce an adaptor-ligated target sequence by ligating an adaptor oligonucleotide to the 5' and 3' ends of an amplified product and amplifying the adaptor-ligated target sequence (see paragraph [0011] and claims 16 and 27). Bradbury *et al.,* also teach that the 5' adaptor is a duplex adaptor or a hairpin adaptor as recited in claim 13, the 5' adaptor is a next generation sequencing (NGS) platform-specific adaptor, an adaptor that contains a single nucleotide overhang, a Y structure or hairpin adaptor with a cleavable site, a fully complementary double-stranded DNA (dsDNA) adaptor, or a dsDNA adaptor with a single-stranded DNA (ssDNA) overhang at the opposite end from the ligation junction as recited in claim 15, and the 5' adaptor comprises at least one of a sequencing primer site, a library amplification primer site, a unique sample identifier (eg., GFP sequence) and a unique molecule identifier sequence as recited in claim 17 (see paragraph [0131] and Table 2).

Therefore, it would have been *prima facie* obvious to one having ordinary skill in the art at the time the invention was made to have performed the method recited in claims 1 and 18 by ligating a 5' adaptor to the 5' end of the target sequence at the blunt or staggered end of step (d) of claim 1 to produce an adaptor-ligated target sequence and amplifying the adaptor-ligated target sequence in view of the prior arts of Shapero *et al.,* and Bradbury *et al.*. One having

ordinary skill in the art would have been motivated to do so because Bradbury *et al.,* have successfully produced an adaptor-ligated target sequence by ligating an adaptor oligonucleotide to the 5' and 3' ends of an amplified product and amplified the adaptor-ligated target sequence (see paragraph [0011] and claims 16 and 27). One having ordinary skill in the art at the time the invention was made would have a reasonable expectation of success to perform the methods recited in claims 1 and 18 by produce an adaptor-ligated target sequence by ligating a 5' adaptor to the 5' end of the target sequence at the blunt or staggered end of step (d) of claim 1 and amplify the adaptor-ligated target sequence in view of the prior arts of Shapero *et al.,* and Bradbury *et al.,* in order to generate a large amount of the adaptor-ligated target sequence.

7. Claims 19 and 21 are rejected under 35 U.S.C. 103 as being unpatentable over Shapero *et al.,* in view of Bradbury *et al.*, as applied to claims 1, 3, 4, 6, 10, 13, 15, 17, 18, and 20 above, and further in view of Robins (US 2014/0322716 A1, priority date: June 15, 2012).

The teachings of Shapero *et al.,* and Bradbury *et al.*, have been summarized previously, *supra.*

Shapero *et al.,* and Bradbury *et al.*, do not disclose that the amplification step is performed using primers that add a sequencing platform-specific sequence to the adaptor ligated target as recited in claim 19. However, Shapero *et al.,* teach quantifying and/or sequencing the target sequence as recited in claim 21 (see claim 26).

Robins teaches that the second amplification primer set 120a, 120b can introduce sequencing platform-specific oligonucleotide sequences to an amplification product in an amplification reaction (see paragraph [0079] and Figure 1).

Therefore, it would have been *prima facie* obvious to one having ordinary skill in the art at the time the invention was made to have performed the methods recited in claim 19 by amplifying an adaptor ligated target sequence using primers that add a sequencing platform-specific sequence to the adaptor ligated target sequence in view of the prior arts of Shapero *et al.*, Bradbury *et al.*, and Robins. One having ordinary skill in the art would have been motivated to do so because Robins has shown that the second amplification primer set 120a, 120b can introduce sequencing platform-specific oligonucleotide sequences to an amplification product in an amplification reaction (see paragraph [0079] and Figure 1). One having ordinary skill in the art at the time the invention was made would have a reasonable expectation of success to perform the method recited in claim 19 by amplifying the adaptor ligated target sequence using primers that can introduce sequencing platform-specific oligonucleotide sequences to an amplification product taught by Robins so that a sequencing platform-specific sequence would be added to the adaptor ligated target sequence in view of the prior arts of Shapero *et al.*, Bradbury *et al.*, and Robins in order to add a sequencing platform-specific sequence to the adaptor ligated target sequence.

Response to Arguments

8. Applicant's arguments with respect to claims 1,3-6,8,10 and 12-21 have been considered but are moot because the arguments do not apply to the current rejection.

Conclusion

9. No claim is allowed.

10. Papers related to this application may be submitted to Group 1600 by facsimile transmission. Papers should be faxed to Group 1600 via the PTO Fax Center. The faxing of such papers must conform with the notices published in the Official Gazette, 1096 OG 30 (November 15, 1988), 1156 OG 61 (November 16, 1993), and 1157 OG 94 (December 28, 1993)(See 37 CAR § 1.6(d)). The CM Fax Center number is (571)273-8300.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Frank Lu, Ph.D., whose telephone number is (571)272-0746. The examiner can normally be reached on Monday-Friday from 9 A.M. to 5 P.M.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Dave Nguyen, can be reached on (571)272-0731.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Frank W Lu /
Primary Examiner, Art Unit 1634
January 22, 2018